Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous Institute Affiliated to University of Mumbai)

Special Examination
July 2024

Max. Marks: 100 Duration: 3Hrs.
Class: T.E. Semester-V Computer Engineering Course Code: CS301
Name of the Course: Software Engineering

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO
Q.1 A)	With respect to rapid application development (RAD) model, answer the following 1. Explain the different life cycle activities that are carried out in the RAD model 2. How does RAD model help accommodate change requests late in the development 3. How does RAD help in faster software development 4. Give examples of two projects for which RAD would be a suitable model for development	 4 2 2 2	CO1
Q.1 B)	Prepare software requirements specification (SRS) document for the automation of examination section of an academic institute.	10	CO1
Q.2 A)	What is Boundary, Controller and Entity object. Draw sequence Diagram for Play move use case in Tic-Tac-Toe	10	CO2
Q.2 B)	What are the different types of cohesion? Give examples of each.	10	CO3
Q.3 A)	What do you mean by balancing a Data Flow Diagram? Draw Data flow diagram (level 0,1 and 2) for Railway Reservation System.	10	CO3
Q.3 B)	How is the software configuration management activity carried out?	10	CO3
Q.4 A)	Explain Model-View-Controller and Publish-Subscribe design pattern	10	CO3
Q.4 B)	Suppose you are the project manager of a software project requiring the following activities	10	CO3

Task Name	Task	Duration	Precedence
A	Specification	15	--
B	Design Database	45	A
C	Design GUI	30	A
D	Code database	105	B
E	Code GUI Part	45	C
F	Integrate and Test	120	D,E
G	Write user manual	60	E,F

a) Determine Earliest Start (ES), Earliest Finish time (EF), Latest start time (LS), Latest finish (LF) for every task
b) Show Critical Path and Compute Slack time (ST) for each task.

Q.5 A)	Consider the following function. Draw Control flow graph. From the control flow graph, determine its Cyclomatic complexity and Find linearly independent paths. Design a test cases for the function.
	
	int compute_gcd(int i, int j,)
	{
	while (i!=j)
	{
	if (i>j) then
	i = i-j;
	else j=j-i;
	}
	return i;
	}
		10	CO4

Q. 5 B)	Discuss DevOps lifecycle and how does it work?	10	CO5

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